BACHELOR OF COMPUTER APPLICATION

SEMESTER 1

FUNDAMENTALS OF MATHEMATICS

Unit 14

Definite Integral and Reduction Formula

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1. INTRODUCTION

In Unit 13, you mastered the techniques of Integration by Parts and Integration of Rational Functions using Partial Fractions. You learned how to tackle integrals involving the product of two functions where one function can be easily differentiated, and the other can be easily integrated. Additionally, you explored how to decompose complex rational functions into simpler fractions that can be integrated individually. These techniques are essential for solving more advanced integration problems and form a strong foundation for the concepts we'll cover in Unit 14.

Unit 14 dives into the world of Definite Integrals and Reduction Formulas. This unit is crucial as it builds on your existing knowledge of integrals and introduces you to the practical applications of definite integrals. You'll learn how to compute definite integrals, which represent the accumulation of quantities and the area under curves. We'll explore the Fundamental Theorem of Calculus, which links differentiation and integration, providing a powerful tool for evaluating definite integrals. You'll also learn about reduction formulas, which are essential for simplifying and evaluating more complex integrals. These concepts are vital for applications in mathematics, physics, engineering, and other fields.

To effectively learn this unit, start by reviewing your understanding of indefinite integrals and the techniques you have mastered so far. Pay close attention to the Fundamental Theorem of Calculus as it forms the basis for evaluating definite integrals. Practice solving problems involving definite integrals to gain confidence in their computation and interpretation. When learning reduction formulas, focus on the step-by-step process of deriving and applying these formulas. Use the examples provided to reinforce your understanding and practice deriving and using reduction formulas on various integrals. Engaging with these concepts through practice and application will help you master the content of Unit 14.

1.1. Learning Objectives

At the end of this unit, you will be able to:

- *Define the term "definite integral" and describe its notation.*
- *Explain the Fundamental Theorem of Calculus and its significance in computing definite integrals.*
- *Calculate definite integrals using antiderivatives and evaluate them at specified limits.*
- *Break down the properties of definite integrals and use them to simplify complex integrals.*
- *Use definite integrals to find the area under simple curves, such as lines, parabolas, circles, and ellipses.*
- *Differentiate between indefinite and definite integrals and identify scenarios where each type is used.*

2. DEFINITE INTEGRAL AND THEIR PROPERTIES

Definite integrals play a fundamental role in calculus, representing the accumulation of quantities and the area under curves. This guide provides an overview of definite integrals, their computation, and essential properties.

2.1. Understanding Definite Integrals

A definite integral of a function $f(x)$ over the interval $[a, b]$ is denoted by:

$$\int_a^b f(x)\,\mathrm{d}x$$

The definite integral represents the net area under the curve of $\int (x)$ from $x = a$ to $x = b$. If $f(x)$ is positive over the interval, the definite integral corresponds to the area under the curve. If $f(x)$ is negative, it represents the negative area.

2.2.1. Fundamental Theorem of Calculus

The Fundamental Theorem of Calculus links differentiation and integration and provides a method for evaluating definite integrals.

Part 1: The Derivative of the Integral

If f is continuous on $[a, b]$ and $F(x) = \int_a^x f(t)\,dt$, then F is differentiable on (a, b), and its derivative id f:

$$\frac{d}{dx}\left(\int_a^x f(t)\,dt\right) = f(x)$$

This part of the theorem shows that integration and differentiation are inverse processes.

Part 2: The Evaluation of the definite integral

If f is continuous on $[a, b]$ and F is an antiderivative of f on [a,b], then:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

This part of the theorem provides a practical way to evaluate definite integrals using antiderivatives.

2.1.2. Computation of Definite Integrals

To compute a definite integral, follow these steps:

Find the antiderivative $F(x)$ of $f(x)$.

Evaluate $F(x)$ at the upper limit b and the lower limit a.

Subtract $F(a)$ from $F(b)$:

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Example:

Evaluate: $\int_1^4 (2x+3)\,dx$

Solution:

Find the antiderivative:

$$\int (2x+3)\,dx = x^2 + 3x + c$$

Evaluate at the limits:

$$F(x) = x^2 + 3x$$

$$F(4) = 4^2 + 3(4) = 16 + 12 = 28$$

$$F(1) = 1^2 + 3(1) = 1 + 3 = 4$$

Subtract $F(a)$ from $F(b)$:

$$\int_j^4 (2x+3)\,dx = 28 - 4 = 24$$

2.2. Properties of Definite Integrals

Definite integrals possess several important properties that facilitate their evaluation and interpretation.

Property 1: Additivity Over Intervals

If $a < c < b$, then:

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$$

Example:

Given $\int_0^2 f(x)\,dx = 5$ and $\int_2^4 f(x)\,dx = 7$, find $\int_0^4 f(x)\,dx$

Solution:

$$\int_0^h f(x)\,dx = \int_0^2 f(x)\,dx + \int_2^4 f(x)\,dx = 5 + 7 = 12$$

Property 2: Linearity

For any constants c_1 and c_2:

$$\int_a^b [c_1 f(x) + c_2 g(x)]\,dx \;=\; c_1 \int_a^b f(x)\,dx + c_2 \int_a^b g(x)\,dx$$

Example:

Evaluate $\int_0^3 [2f(x) + 3g(x)]\,dx$, given $\int_0^3 f(x)\,dx = 4$ and $\int_0^3 g(x)\,dx = 5$

Solution:

$$\int_0^3 [2f(x) + 3g(x)]\,dx \;=\; 2\int_0^3 f(x)\,dx + 3\int_0^3 g(x)\,dx = 2(4) + 3(5) = 8 + 15 = 23$$

Property 3: Zero Width interval

If the limits of integration are the same, the definite integral is zero:

$$\int_a^a f(x)\,dx = 0$$

Property 4: Reversal of Limits

Reversing the limits of integration changes the sign of the definite integral:

$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

Property 5: Non-negativity

If $f(x) \geq 0$ for all x in $[a, b]$, then:

$$\int_o^b f(x)\,dx \geq 0$$

Property 6: Comparison

If $f(x) \leq g(x)$ for all x in $[a, b]$, then:

$$\int_a^b f(x)\,dx \le \int_a^b g(x)\,dx$$

Examples:

Problem 1: Use properties to evaluate $\int_1^3 (4x^2 - 2x)dx$.

Solution:

1. Split the integral:

$$\int_1^3 (4x^2 - 2x)dx = \int_1^3 4x^2 dx - \int_1^3 2x dx$$

2. Use the linearity property:

$$\int_1^3 4x^2 dx = 4\int_1^3 x^2 dx$$
$$\int_1^3 2x dx = 2\int_1^3 x dx$$

3. Compute each integral separately:

$$\int_1^3 x^2 dx = \left[\frac{x^3}{3}\right]_{-1}^3 = \frac{27}{3} - \frac{1}{3} = 9 - \frac{1}{3} = \frac{26}{3}$$
$$\int_1^3 x dx = \left[\frac{x^2}{2}\right]_1^3 = \frac{9}{2} - \frac{1}{2} = \frac{8}{2} = 4$$

4. Combine the results:

$$\int_1^3 (4x^2 - 2x)dx = 4 \cdot \frac{26}{3} - 2 \cdot 4$$
$$= \frac{104}{3} - 8 = \frac{104}{3} - \frac{24}{3} = \frac{80}{3}$$

So, $\int_1^3 (4x^2 - 2x)dx = \frac{80}{3}$.

Problem 2: Given $\int_0^2 f(x)dx = 3$ and $\int_2^4 f(x)dx = -1$, find $\int_0^4 f(x)dx$.

Solution:

Using the additivity property:

$$\int_0^4 f(x)dx = \int_0^2 f(x)dx + \int_2^4 f(x)dx$$
$$= 3 + (-1) = 2$$

Definite integrals are essential tools in calculus for measuring accumulated quantities and areas under curves. By understanding their properties and learning to compute them effectively, one can solve a wide range of mathematical and real-world problems. These properties not only simplify the computation of definite integrals but also provide deeper insights into the behaviour of functions over specified intervals.

3. REDUCTION FORMULA

Reduction formulas are powerful tools in calculus used to simplify the process of evaluating integrals, especially when dealing with complex or higher-order integrals. These formulas help reduce an integral to a simpler form, often in terms of another integral of a lower order.

Understanding Reduction Formulas

Reduction formulas are recursive relationships that express a given integral in terms of another integral with a reduced power or simpler integrand. This method is particularly useful for integrals involving powers of trigonometric, exponential, and polynomial functions.

Deriving Reduction Formulas

To derive a reduction formula, integration by parts is often employed. The general approach is to choose parts of the integrand that, when integrated and differentiated, simplify the original integral into a form that includes an integral of a lower power or simpler function.

3.1. Reduction Formula for Powers of Sine

Consider the integral $I_n = \int sin^n(x)\,dx$, where n is a positive integer. We can derive a reduction formula for I_n using integration by parts.

Integration by Parts

Let $u = sin^{n-1}(x)$ and $dv = sin(x)\,dx$. Then, $du = (n-1)\,sin^{n-2}(x)\,cos(x)\,dx$ and $v = -cos(x)$

Using Integration by parts:

$$I_n = \int sin^n(x)\,dx = \int sin^{n-1}(x)\,sin(x)\,dx$$

$$\int u\,dv = uv - \int v\,du$$

$$I_n = -sin^{n-1}(x)\,cos(x) - \int (-cos(x))(n-1)\,sin^{n-2}(x)\,cos(x)\,dx$$

$$I_n = -sin^{n-1}(x)\,cos(x) + (n-1)\int sin^{n-2}(x)\,cos^2(x)\,dx$$

Simplifying:

Use the identity $cos^2(x) = 1 - sin^2(x)$:

$$I_n = -sin^{n-1}(x)\,cos(x) + (n-1)\int sin^{n-2}(x)(1 - sin^2(x))\,dx$$

$$ln = -sin^{n-1}(x)\,cos(x) + (n-1)\int sin^{n-2}(x)\,dx - (n-1)\int sin^n(x)\,dx$$

Formulating the reduction:

$$I_n = -\sin^{n-1}(x)\cos(x) + (n-1)I_{n-2} - (n-1)I_n$$
$$I_n + (n-1)I_n = \sin^{n-1}(x)\cos(x) + (n-1)I_{n-2}$$
$$nI_n = -\sin^{n-1}(x)\cos(x) + (n-1)I_{n-2}$$
$$I_n = \frac{-\sin^{n-1}(x)\cos(x)}{n} + \frac{n-1}{n}I_{n-2}$$

Thus, the reduction formula for $I_n = \int \sin^n(x)\,dx$ is:

$$I_n = \int \sin^n(x)\,dx = \frac{-\sin^{n-1}(x)\cos(x)}{n} + \frac{n-1}{n}\int \sin^{n-2}(x)\,dx$$

3.2. Reduction Formula for Powers of Cosine

Similarly, we ca derive a reduction formula for $J_n = \int \cos^n(x)\,dx$

Integration by parts:

Let $u = \cos^{n-1}(x)$ and $dv = \cos(x)\,dx$. Then, $du = -(n-1)\cos^{n-2}(x)\sin(x)\,dx$ and $v = \sin(x)$.

Using Integration by parts:

$$J_n = \int \cos^n(x)\,dx = \int \cos^{n-1}(x)\cos(x)\,dx$$
$$\int u\,dv = uv - \int v\,du$$
$$J_n = \cos^{n-1}(x)\sin(x) - \int \sin(x)\big(-(n-1)\big)\cos^{n-2}(x)\sin(x)\,dx$$
$$J_n = \cos^{n-1}(x)\sin(x) + (n-1)\int \cos^{n-2}(x)\sin^2(x)\,dx$$

Simplifying:

Use the identity

$$\sin^2(x) = 1 - \cos^2(x)$$
$$J_n = \cos^{n-1}(x)\sin(x) + (n-1)\int \cos^{n-2}(x)(1-\cos^2(x))\,dx$$
$$J_n = \cos^{n-1}(x)\sin(x) + (n-1)\int \cos^{n-2}(x)\,dx - (n-1)\int \cos^n(x)\,dx$$

Formulating the reduction:

$$J_n = \cos^{n-1}(x)\sin(x) + (n-1)J_{n-2} - (n-1)J_n$$
$$J_n + (n-1)J_n = \cos^{n-1}(x)\sin(x) + (n-1)J_{n-2}$$
$$nJ_n = \cos^{n-1}(x)\sin(x) + (n-1)J_{n-2}$$
$$J_n = \frac{\cos^{n-1}(x)\sin(x)}{n} + \frac{n-1}{n}J_{n-2}$$

Thus, the reduction formula for $J_n = \int \cos^n(x)\,dx$ is:

$$J_n = \int \cos^n(x)\,dx = \frac{\cos^{n-1}(x)\sin(x)}{n} + \frac{n-1}{n}\int \cos^{n-2}(x)\,dx$$

3.3. Reduction Formula for $x^n e^x$

For integrals involving products of polynomials and exponentials, such as $\int x^n e^x\,dx$, we can derive a reduction formula as follows:

Integration by parts:

Let, $u = x^n$ and $dv = e^x\,dx$. Then,, $du = nx^{n-1}$ and $v = e^x$.

Using Integration by parts:

$$\int x^n e^x\,dx = x^n e^x - \int nx^{n-1} e^x\,dx$$

Formulating the Reduction:

$$I_n = x^n e^x - n\int x^{n-1} e^x\,dx$$

$$I_n = x^n e^x - nI_{n-1}$$

Thus, the reduction formula for $I_n = \int x^n e^x\,dx$ is:

$$I_n = \int x^n e^x\,dx = x^n e^x - n\int x^{n-1} e^x\,dx$$

$$I_n = x^n e^x - nI_{n-1}$$

Reduction formulas are essential tools for evaluating complex integrals, especially those involving powers of functions or products of different types of functions. By using integration by parts and simplifying the integrand, these formulas reduce higher-order integrals to simpler forms that are easier to evaluate. Mastery of reduction formulas allows for efficient and accurate computation of a wide range of integrals in calculus.

4. DEFINITE INTEGRALS AS A LIMIT OF SUM

The definite integral of a function over an interval $[a,b]$ can be understood as the limit of a sum of areas of rectangles. This approach is the basis of Riemann sums and provides an intuitive understanding of integration.

4.1. Riemann Sums

To approximate the definite integral of a function $f(x)$ over the interval $[a, b]$, we divide the interval into n subintervals of equal width $\Delta x = \frac{b-a}{n}$. For each sub-interval, we choose a sample point x_i^* within the sub-interval $[x_{i-1}, x_i]$. The Riemann sum is the sum of the areas of rectangles with heights $f(x_i^*)$ and widths Δx:

$$S_n = \sum_{i=1}^{n} f(x_i^*)\Delta x$$

As n approaches infinity, the width of the subintervals Δx approaches zero, and the Riemann sum approaches the exact value of the definite integral:

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \sum_{i=1}^{n} f(x_i^*)\Delta x$$

Example:

Approximate $\int_0^1 x^2\,dx$ using the limit of a Reimann sum

Solution:

Divide the interval [0, 1] into n subintervals of equal width $\Delta x = \frac{1-0}{n} = \frac{1}{n}$

Choose sample points $x_i^* = \frac{i}{n}$ for $i = 1,2, \dots, n$

Form the Reimann sum:

$$S_n = \sum_{i=1}^{n} \left(\frac{i}{n}\right)^2 \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^{n} i^2$$

Use the formula for sum of squares:

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$$

Substitute and simplify:

$$S_n = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{(n+1)(2n+1)}{6n^2}$$

Take the limit as $n \to \infty$:

$$\lim_{n\to\infty} \frac{(n+1)(2n+1)}{6n^2} = \frac{2}{6} = \frac{1}{3}$$

So, $\int_0^1 x^2\, dx = \frac{1}{3}$

Definite integrals as a limit of a sum provide an intuitive understanding of integration by approximating the area under a curve using rectangles. Mastery of this concepts is essential for solving a wide range of problems in calculus and its applications in science, engineering, and other fields.

5. EVALUATION OF DEFINITE INTEGRAL

To evaluate a definite integral, follow these steps:

1. Find the Antiderivative: Determine the antiderivative $F(x)$ of the integrand $f(x)$
2. Evaluate at the Limits: Compute $F(b) - F(a)$

1. Evaluate $\int_1^3 (3x^2 - 2x + 1)\, dx$

Solution:

$$\int (3x^2 - 2x + 1)\, dx = x^3 - x^2 + x + c$$

Evaluate at the limits:

$$[x^3 - x^2 + 1]_1^3 = (3^3 - 3^2 + 1) - (1^3 - 1^2 + 1) = (27 - 9 + 3) - (1 - 1 + 1) = 21 - 1$$
$$= 20$$

2. Evaluate $\int_0^1 e^x\, dx$

Solution:

$$\int e^x\, dx = e^x + c$$

Evaluate at the limits:

$$[e^x]_0^1 = e^1 - e^o = e - 1$$

3. Evaluate $\int_0^{\pi/2} sin(x)\, dx$

Solution:

$$\int sin(x)\, dx = -cos(x) + c$$

Evaluate at the limits:

$$[-cos(x)]_0^{\pi/2} = -cos(\pi / 2) - (-cos(0)) = -0 - (-1) = 1$$

4. Evaluate $\int_1^2 \frac{1}{x} dx$

Solution:

$$\int \frac{1}{x} dx = ln|x| + c$$

Evaluate at the limits:

$$[ln|x|]_1^2 = \ln(2) - \ln(1) = \ln(2) - 0 = \ln(2)$$

Understanding the basic properties of definite integrals simplifies their evaluation and provides a deeper insight into their behavior. These properties, combined with the Fundamental Theorem of Calculus, make evaluating definite integrals straightforward and systematic. Mastery of these techniques and properties is essential for solving a wide range of problems in calculus and its applications in science, engineering, and other fields.

INSPIRED BY LIFE

6. APPLICATIONS OF DEFINITE INTEGRALS IN FINDING THE AREA UNDER SIMPLE CURVES

Definite integrals are a powerful tool for finding the area under curves, which has numerous applications in mathematics and science. This guide explores how to use definite integrals to find the area under simple curves, including lines, circles, parabolas, and ellipses in their standard forms.

6.1. Finding the Area Under a Line

The area under a linear function can be calculated using definite integrals. Consider a line given by $y = mx + b$

Find the area under the line y=2x+3 from x = 1 to x = 4.

Solution:

Set up the definite integral:

$$\int_1^4 (2x + 3)\, dx$$

Find the antiderivative(integral):

$$\int (2x + 3)\, dx = x^2 + 3x + c$$

Evaluate at the limits:

$$[x^2 + 3x]_1^4 = [4^2 + 3(4)] - [1^2 + 3(1)] = (16 + 12) - (1 + 3) = 28 - 4 = 24$$

Therefore, the area under the curve $y = 2x + 3$ from x = 1 to x = 4 is 24 square units.

6.2. Finding the Area Under a Parabola

The area under a parabolic curve can also be calculated using definite integrals. Consider a parabola given by $y = ax^2 + bx + c$

Find the areas under the parabola $y = x^2$ from x = 0 to x = 2

Solution:

$$\int_0^2 x^2\, d\alpha = \frac{x^3}{3} + c$$

Evaluating at the limits:

$$\left[\frac{x^3}{3}\right]_0^2 = \frac{2^3}{3} - \frac{0^3}{3} = \frac{8}{3} - 0 = \frac{8}{3}$$

Therefore, the area under the parabola $y = x^2$ for x = 0 and x = 2 is $\frac{8}{3}$ square units.

6.3. Finding the Area Under a Circle

For circles, the standard form equation is $x^2 + y^2 = r^2$. The area can be found by integrating the upper or lower semicircle and then doubling the result.

Find the area under the upper semicircle of the circle $x^2 + y^2 = 4$ (i.e. $y = \sqrt{4 - x^2}$) from x = -2 to x = 2.

Solution:

Set up the definite integral for the upper semicircle:

$$\int_{-2}^{2} \sqrt{4 - x^2}\, dx$$

Use the symmetry of the circle:

The integral can be computed over [0,2] and then doubled:

$$2\int_{0}^{2} \sqrt{4 - x^2}\, dx$$

Using trigonometric substitution:

Let $x = 2\, sin(\theta)$, hence $dx = 2\, cos(\theta)\, d\theta$

When x = 0, and $\theta = 0$, and when x = 2, $\theta = \frac{\pi}{2}$

$$I_n = 2\int_{0}^{\pi/2} \sqrt{4 - 4\, sin^2(\theta)} \cdot 2\, cos(\theta)\, d\theta$$

$$I_n = 4\int_{0}^{\pi/2} \sqrt{4\, cos^2(\theta)} \cdot cos(\theta)\, d\theta$$

$$I_n = 4\int_{0}^{\pi/2} 2\, cos^2(\theta)\, d\theta$$

$$I_n = 8\int_0^{\frac{\pi}{2}} cos^2(\theta)\, d\theta$$

Use the identity $cos^2(\theta) = \frac{1+cos(2\theta)}{2}$:

Therefore,

$$I_n = 8\int_0^{\frac{\pi}{2}} \frac{1+cos(2\theta)}{2} d\theta$$

$$I_n = 4\int_0^{\frac{\pi}{2}} 1 + cos(2\theta)\, d\theta$$

$$I_n = 4\left[\theta + \frac{sin(2\theta)}{2}\right]_0^{\frac{\pi}{2}}$$

$$I_n = 4\left[\frac{\pi}{2} + 0\right] = 2\pi$$

Therefore, the area under the upper semicircle of the circle given is 2π square units

6.4. Finding the Area Under an Ellipse

For ellipses, the standard form equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The area can be found by integrating the upper or lower half and then doubling the result.

Find the area under the upper half of the ellipse $\frac{x^2}{4} + \frac{y^2}{1} = 1$ (i.e. $y = \sqrt{1 - \frac{x^2}{4}}$) from x = -2 to x = 2.

Solution:

Set up the definite integral for the upper half:

$$\int_{-2}^{2} \sqrt{1 - \frac{x^2}{4}} dx$$

Use the symmetry of the ellipse:

The integral can be computed over [0,2] and then doubled:

$$I_n = 2\int_0^2 \sqrt{1-\frac{x^2}{4}}\,dx$$

Using trigonometric substitution:

Let $x = 2\,sin(\theta)$

Therefore $dx = 2\,cos(\theta)\,d\theta$.

When x = 0, $\theta = 0$ and x = 2, $\theta = \frac{\pi}{2}$

$$I_n = 2\int_0^{\frac{\pi}{2}} \sqrt{1-sin^2\,\theta}\cdot 2\,cos(\theta)\,d\theta = 4\int_0^{\pi/2} cos^2(\theta)\,d\theta$$

Using the identity:

$$cos^2(\theta) = 1 + \frac{cos(2\theta)}{2}$$

$$I_n = 4\int_0^{\frac{\pi}{2}} \frac{1+cos(2\theta)}{2}\,d\theta$$

$$I_n = 2\int_0^{\frac{\pi}{2}} 1 + cos(2\theta)\,d\theta$$

$$I_n = 2\left[\theta + \frac{sin(2\theta)}{2}\right]_0^{\pi/2} = 2\left[\frac{\pi}{2}+0\right] = \pi$$

So, the area under the upper half of the ellipse given is π square units.

Definite integrals are a powerful tool for calculating the area under simple curves, including lines, parabolas, circles, and ellipses. By setting up the appropriate integral and using the properties of definite integrals, one can find the exact area under these curves. Mastery of these techniques is essential for solving a wide range of problems in calculus and its applications in science, engineering, and other fields.

7. WORKING EXAMPLES

1. Evaluate $I = \int_1^2 (4x^3 - 5x^2 + 6x + 9)dx$

   Solution:

   We know

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

   Consider

$$I = \int_1^2 (4x^3 - 5x^2 + 6x + 9)dx$$

$$= \left[x^4 - \frac{5x^3}{3} + 3x^2 + 9x\right]_1^2 = \left[16 - \frac{40}{3} + 12 + 18 - 1 + \frac{5}{3} - 3 - 9\right] = \frac{64}{3}$$

2. Evaluate: $I = \int_0^1 \frac{1}{1+x^2} dx$

   Solution:

   We know

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

   Consider

$$I = \int_0^1 \frac{1}{1+x^2} dx \quad = [\tan^{-1} x]_0^1 = \tan^{-1} 1 - tan^{-1} 0 = \frac{\pi}{4}$$

3. Evaluate $I = \int_{-1}^1 (x+1)dx$

   Solution:

   We know

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

   Consider

$$I = \int_{-1}^1 (x+1)dx \quad = \left[\frac{x^2}{2} + x\right]_{-1}^1 = \left[\frac{1}{2} + 1 - \frac{1}{2} + 1\right] = 2$$

4. Evaluate $I = \int_0^{\frac{\pi}{4}} (2\sec^2 x + x + 2)dx$

Solution:

We know

$$\int_a^b f(x)dx = [F(x)]_a^b = F(b) - F(a)$$

Consider

$$I = \int_0^{\frac{\pi}{4}} (2\sec^2 x + x + 2)dx = \left[2\tan x + \frac{x^2}{2} + 2x\right]_0^{\frac{\pi}{4}}$$

$$= 2\tan\left(\frac{\pi}{4}\right) + \frac{\left(\frac{\pi}{2}\right)^2}{2} + 2\left(\frac{\pi}{4}\right) = 2 + \frac{\pi^2}{8} + \frac{\pi}{2}$$

5. Evaluate $Area = 4\int_0^a y\,dx$

Solution:

But $y = \frac{b}{a}\sqrt{a^2 - x^2}$

Thus,

$$Area = 4\int_0^a y\,dx = 4\int_0^a \frac{b}{a}\sqrt{a^2 - x^2}\,dx$$

We use the formula:

$$\int \sqrt{a^2 - x^2}dx = \frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)$$

Thus,

$$Area = \frac{4b}{a}\left[\frac{x\sqrt{a^2 - x^2}}{2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)\right]_0^a = \frac{4b}{a}\left[\frac{a^2}{4}\pi\right] = \pi ab\ square\ units.$$

8. SUMMARY

Congratulations on reaching the end of Unit 14! In this unit, we delved into the fascinating world of definite integrals and reduction formulas. We started by revisiting what we learned in Unit 13, where we explored Integration by Parts and the Integration of Rational Functions using Partial Fractions. Building on this foundation, we shifted our focus to the powerful concept of definite integrals.

Definite integrals play a crucial role in calculus as they allow us to measure the accumulation of quantities, which can be visualized as the area under curves. We began by understanding the notation and the fundamental theorem of calculus, which elegantly links differentiation and integration. This theorem not only provides a method for evaluating definite integrals but also highlights the inverse relationship between these two core concepts of calculus.

We then explored the properties of definite integrals, such as additivity over intervals, linearity, and non-negativity. These properties simplify the computation of definite integrals and offer valuable insights into the behavior of functions over specified intervals. Through practical examples, we demonstrated how to apply these properties to solve problems efficiently. For instance, we learned to split integrals into smaller intervals, factor out constants, and even handle integrals over zero-width intervals.

Moving forward, we tackled the computation of definite integrals by finding antiderivatives and evaluating them at the given limits. We practiced this technique with various functions, reinforcing our understanding through step-by-step examples. From simple polynomials to more complex functions involving trigonometric, exponential, and logarithmic forms, we mastered the art of evaluating definite integrals.

Another exciting topic we covered was the use of definite integrals to find the area under curves. We applied this knowledge to calculate the area under lines, parabolas, circles, and ellipses. This practical application of definite integrals is essential in fields like physics, engineering, and economics, where understanding the area under a curve can provide insights into distance traveled, work done, and economic profits.

Finally, we introduced reduction formulas as powerful tools for simplifying complex integrals. These formulas allow us to reduce higher-order integrals into more manageable

forms, making the evaluation process more straightforward. We derived reduction formulas for various functions, including powers of sine, cosine, and exponential functions, showcasing how integration by parts can be employed to achieve this simplification.

By mastering definite integrals and reduction formulas, you have equipped yourself with essential tools for solving a wide range of mathematical and real-world problems. Keep practicing these techniques, and you'll continue to build a strong foundation in calculus that will serve you well in advanced studies and professional applications. Well done!

INSPIRED BY LIFE

9. SELF-ASSESSMENT QUESTIONS

1. The definite integral of a function $f(x)$ from a to b is represented as $\int_a^b$ _____.
2. What is the Fundamental Theorem of Calculus used for?

   a) Differentiating functions

   b) Evaluating definite integrals

   c) Finding limits

   d) Solving differential equations
3. The formula for integration by parts is $\int u\,dv = uv - \int v$_____.
4. Which of the following properties states that reversing the limits of integration changes the sign of the definite integral?

   a) Additivity Over Intervals

   b) Linearity

   c) Zero Width Interval

   d) Reversal of Limits
5. To compute a definite integral $\int_a^b f(x)dx$, you need to find the antiderivative of $f(x)$ and then evaluate it at the upper and lower limits, which is represented as F(b) - _____.
6. What is the definite integral of $\int_0^2 x^2 dx$?

   a) 4

   b) $\frac{8}{3}$

   c) 2

   d) 6
7. The reduction formula for the integral $I_n = \int \sin^n(x)dx$ is $I_n = \int sin^n(x)\,dx = \frac{-sin^{n-1}(x)\,cos(x)}{n} + \frac{n-1}{n}\int \sin^{n-2}(x)$_____.
8. Which method is used to approximate the definite integral of a function by dividing the interval into subintervals and summing up the areas of rectangles?

   a) Trapezoidal Rule

   b) Simpson's Rule

c) Riemann Sum

d) Midpoint Rule

9. The property of definite integrals that states $\int_a^a f(x)dx = 0$ is known as the _____ Width Interval property.

10. What is the definite integral of $\int_0^1 e^x dx$?

a) $e - 1$

b) e

c) $\frac{e^2}{2}$

d) $1 - e$

10. TERMINAL QUESTIONS

1. Evaluate the definite integral: $\int_0^2 x^2 dx$
2. Find the definite integral: $\int_1^3 (3x^2 - 2x + 1)dx$
3. Evaluate the integral using integration by parts: $\int xe^x dx$
4. Find the antiderivative and then evaluate the definite integral: $\int_0^1 e^x dx$
5. Use a reduction formula to evaluate: $\int \sin^2(x)dx$
6. Compute the definite integral: $\int_0^{\frac{\pi}{2}} \sin(x)dx$
7. Evaluate the definite integral using properties: $\int_1^3 (4x^2 - 2x)dx$
8. Find the integral of the rational function: $\int \frac{2x+3}{(x-1)(x+2)} dx$
9. Evaluate the definite integral: $\int_1^2 \frac{1}{x} dx$
10. Compute the definite integral using the Fundamental Theorem of Calculus: $\int_0^2 (2x + 3)dx$
11. Find the area under the curve: $y = x^2$ from $x = 0$ to $x = 2$
12. Calculate the definite integral: $\int_0^{\pi} \sin^3(x)dx$
13. Evaluate the definite integral using trigonometric substitution: $\int_{-2}^2 \sqrt{4 - x^2} dx$
14. Determine the value of the definite integral: $\int_0^1 \frac{1}{1+x^2} dx$
15. Evaluate the definite integral: $\int_0^2 x^3 dx$
16. Compute the definite integral for the given function: $\int_1^4 (2x + 1)dx$
17. Find the value of the definite integral: $\int_0^1 \cos(x)dx$
18. Evaluate the definite integral using the method of partial fractions: $\int \frac{3x+5}{(x-1)^2(x+1)} dx$
19. Calculate the definite integral using the Fundamental Theorem of Calculus: $\int_1^3 x^2 dx$
20. Determine the value of the integral using integration by parts: $\int x\ln(x)dx$

11. ANSWERS

Self-Assessment Questions

1. $f(x)dx$
2. b) Evaluating definite integrals
3. du
4. d) Reversal of Limits
5. $F(a)$
6. b) $\frac{8}{3}$
7. dx
8. c) Riemann Sum
9. Zero
10. a) $e-1$

Terminal Questions

1. $\int_0^2 x^2 dx = \frac{8}{3}$
2. $\int_1^3 (3x^2 - 2x + 1)dx = 20$
3. $\int xe^x dx = xe^x - e^x + C$
4. $\int_0^1 e^x dx = e - 1$
5. $\int \sin^2(x)dx = \frac{x}{2} - \frac{\sin(2x)}{4} + C$
6. $\int_0^{\frac{\pi}{2}} \sin(x)dx = 1$
7. $\int_1^3 (4x^2 - 2x)dx = \frac{80}{3}$
8. $\int \frac{2x+3}{(x-1)(x+2)} dx = 2\ln|x-1| - \ln|x+2| + C$
9. $\int_1^2 \frac{1}{x} dx = \ln(2)$
10. $\int_0^2 (2x+3)dx = 10$
11. Area under $y = x^2$ from $x = 0$ to $x = 2$ is $\frac{8}{3}$ square units.

12. $\int_0^{\pi} \sin^3(x)dx = 0$

13. $\int_{-2}^{2} \sqrt{4-x^2}dx = 2\pi$

14. $\int_0^1 \frac{1}{1+x^2}dx = \frac{\pi}{4}$

15. $\int_0^2 x^3 dx = 4$

16. $\int_1^4 (2x+1)dx = 21$

17. $\int_0^1 \cos(x)dx = \sin(1)$

18. $\int \frac{3x+5}{(x-1)^2(x+1)}dx = -4\ln|x-1| + 4\ln|x+1| + \frac{4}{x-1} + C$

19. $\int_1^3 x^2 dx = \frac{26}{3}$

20. $\int x\ln(x)dx = \frac{x^2}{2}\ln(x) - \frac{x^2}{4} + C$